



6. (a)

$$(a) P = (F87B)_{16} = 1111100001111011$$

$$2\text{'s complement} = 1111100001111011$$

$$\begin{array}{r} 11110000101 \\ +1 \\ \hline (0000 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1)_2 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ 1024 \quad 512 \quad 256 \quad 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \end{array}$$

Value in decimal is $(1925)_{10}$, Hence, $P = -1925$

So, now doing $8 * P = (-15400)_{10}$

Since, a negative number we have to find the 2 's complement of -15400

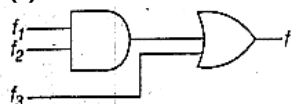
Binary of 15400 = 0011110000101000

$$2\text{'s complement} = 1100 \quad 0011 \quad 1101 \quad 1000$$

$$\begin{array}{cccc} \boxed{} & \boxed{} & \boxed{} & \boxed{} \\ C & 3 & D & 8 \end{array}$$

$$= (C3 D8)_{16}$$

7. (c)



From the diagram

$$f = f_1 f_2 + f_3 \text{ (put value)}$$

$$= (4,5,6,7,8)f_2 + (1,6,15) = (1,6,8,15)$$

Now, as you know that the number 0,1,2, ... 14,15 represent the minterms like $\Sigma_m(6,8)$.

- (i) If you AND (4,6) with (4,5,6,7,8) it will result in (4,6) but we don't need 4, so option (a) is wrong.
- (ii) If you AND (4,8) with (4,5,6,7,8) it will result in (4,8) but we don't need 4, so option (b) is wrong.
- (iii) If you AND (4,6,8) with (4,5,6,7,8) it will result in (4,6,8) but we don't need 4 here too, so option (d) is wrong.
- (iv) If you AND (6,8) with (4,5,6,7,8) it will result in (6,8) now when OR(6,8) with (1,6,15), you get (1,6,8,15) which is f , so option (c) is correct

8. (d)

Given, expression $\overline{(x + y + z)}$

As, we know De-morgan's law says

$$\overline{a + b} = \bar{a} \cdot \bar{b}$$

So, let

$$x + y = a$$

$$z = b$$

$$\Rightarrow \overline{(x + y + z)} = \overline{x + y} \cdot \bar{z}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\Rightarrow \overline{x + y} \cdot \bar{z} = (x + y) \cdot z$$

$$\text{Hence, } \overline{x + y + z} = (x + y) \cdot z$$

9. (c) Since, $q \rightarrow s$ and $r \rightarrow s$, thus

$(q \vee r) \rightarrow s, p \rightarrow (q \vee r)$ is given, then by rule of 'Syllogism'. $p \rightarrow s$ is deduced.

10. (a) Dynamic random access memories (DRAMs) are semiconductor integrated circuits (ICs) that operate like a bank of capacitors. DRAM consist atleast 1 MOS transistor necessarily.